

EART11200 - The Natural Scientist's Toolkit
Mathematics and Physics

Week 3 Lecture - Differentiating Combinations of Functions

Last Week...

- For a function, $y = f(x)$, that describes some process we are interested in, in one dimension, the derivative:

$$f'(x) = \frac{dy}{dx}$$

- It is interpreted as the rate of change of 'y' with respect to 'x'. Lets imagine we are observing this change over a discrete interval, such that we could write this as an approximation:

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$$

LAST WEEK...

- We can apply similar procedures to other common functions and start populating a list of rules.
- The more you practice these procedures, the easier it will become to apply them without looking them up.
- Use your math's survival pack when needed!

Function $y = f(x)$	**Derivative $\frac{dy}{dx}$**
constant	0
x	1
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
$ln(x)$	$\frac{1}{x}$
$sin(x)$	$cos(x)$
$cos(x)$	$-sin(x)$

Last Week...

- What happens when we have a sum of functions?

$$y = 5x + e^x$$

- Well luckily, the derivative of the sum is the sum of the derivatives!
- This means we can just differentiate each term individually

$$\frac{dy}{dx} = 5 + e^x$$

- We'll see next week what happens when we have more complicated combinations of functions.

Function $y = f(x)$	**Derivative $\frac{dy}{dx}$**
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e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
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Course layout

Date	Week	Topic
27/01/2025	20	Calculating Gradients
03/02/2025	21	Introduction to Differentiation
10/02/2025	22	Differentiating Combinations of Functions
17/02/2025	23	Optimisation & Properties of Derivatives
24/02/2025	24	Second Order Derivatives & Root Finding
03/03/2025	25	Introduction to Integration
10/03/2025	26	Integration using Newtons Laws
17/03/2025	27	Integration by Substitution
24/03/2025	28	Integrating Second Order ODEs
31/03/2025	29	Vectors and Partial Differentiation
07/04/2025	30	Non-teaching week
14/04/2025	31	Easter break
21/04/2025	32	Easter break
28/04/2025	33	Revision: Differentiation
05/05/2025	34	Revision: Integration
Exam Period		

Intended Learning Outcomes

(1) Apply the chain rule to differentiate composite functions in theoretical and real-world applications.

(2) Apply the product rule to differentiate products of functions in theoretical and real-world applications.

(3) Apply the quotient rule to differentiate products of functions in theoretical and real-world applications.

Example: Hydrostatic Equation

- Air density depends on pressure, P, and temperature, T, through the ideal gas law, $\rho = \frac{P}{R_\alpha T}$:

$$\frac{dP}{dz} = -\frac{g}{R_\alpha T} P$$

where g and R_α are constants.

- This is a differential equation, and it has solution:

$$P(z) = P_0 \exp\left(-\frac{g}{R_\alpha T} z\right)$$

where P_0 is the pressure at surface level

- We can use differentiation to show that this is true

$$\frac{dP}{dz} = -\frac{g P_0}{R_\alpha T} \exp\left(-\frac{g}{R_\alpha T} z\right) = -\frac{P}{R_\alpha T} P$$

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The Chain Rule

- So far, functions have matched one of the rules listed on the right hand side:

$$f(x) = \sin(x) \qquad \frac{df}{dx} = \cos(x)$$

- However, there are instances where no level of re-arranging will get to a form we can use.
- What happens when we get the following?

$$f(x) = \sin(3x + 1) \qquad \frac{df}{dx} = ???$$

- When we have functions with functions $f(g(x))$, we can use a nice mathematical trick to help here:

$$\frac{df}{dx} = \frac{df}{dx} \times 1 = \frac{df}{dx} \times \frac{dg}{dg} = \frac{df}{dg} \times \frac{dg}{dx}$$

Function $y = f(x)$	**Derivative $\frac{dy}{dx}$**
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The Chain Rule

- This is known as the **Chain Rule**.
- Let's see it in action:

$$f(x) = \sin(3x + 1)$$

- **Step 1**: Split the function and substitute

$$f(g) = \sin(g) \qquad g(x) = 3x + 1$$

- **Step 2**: Differentiate:

$$\frac{df}{dg} = \cos(g) \qquad \frac{dg}{dx} = 3$$

- **Step 3**: Apply the product rule and substitute back in:

$$\frac{df}{dx} = 3\cos(3x + 1)$$

The Chain Rule

$$\frac{df}{dx} = \frac{df}{dg} \times \frac{dg}{dx}$$

Function $y = f(x)$	Derivative $\frac{dy}{dx}$
constant	0
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The Product Rule

- What happens when we get the following?

$$f(x) = x\exp(3x) \qquad \frac{df}{dx} = ???$$

- Let's think of a simpler example:

$$f(x) = u(x)v(x) = x^3 \times x^6 = x^9$$

- Can we just multiply the derivatives?

$$u(x) = x^3 \implies \frac{du}{dx} = 3x^2$$

$$v(x) = x^6 \implies \frac{du}{dx} = 6x^5$$

$$\frac{df}{dx} = \frac{du}{dx} \times \frac{dv}{dx} = (3x^2)(6x^5) = 18x^7$$

Function $y = f(x)$	**Derivative $\frac{dy}{dx}$**
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$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
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The Product Rule

- This is incorrect as from the table of common differentials we know that

$$\frac{df}{dx} = 9x^8$$

- When we have two functions multiplied together, $f(x) = u(x)v(x)$, we can use the **product rule**:

$$\frac{df}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

- This rule allows us to increase the complexity of the functions we study!

Function $y = f(x)$	**Derivative $\frac{dy}{dx}$**
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The Product Rule

- Let's see it in action

$$f(x) = x\exp(3x)$$

- **Step 1**: Split the function:

$$u(x) = x \qquad v(x) = \exp(3x)$$

- **Step 2**: Differentiate:

$$\frac{du}{dx} = 1 \qquad \frac{dv}{dx} = 3\exp(3x)$$

- **Step 3**: Apply the product rule and substitute back in:

$$\frac{df}{dx} = 3x\exp(3x) + \exp(3x) = (3x+1)\exp(3x)$$

The Product Rule

$$\frac{df}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

Function $y = f(x)$	Derivative $\frac{dy}{dx}$
constant	0
x	1
x^n	nx^{n-1}
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The Quotient Rule

- What happens when we get the following?

$$f(x) = \frac{3x+1}{6x-1} \qquad \frac{df}{dx} = ???$$

- Let's think of a simpler example:

$$f(x) = \frac{u(x)}{v(x)} = \frac{x^3}{x^6} = x^{-3}$$

- Can we just divide the derivatives?

$$u(x) = x^3 \implies \frac{du}{dx} = 3x^2$$

$$v(x) = x^6 \implies \frac{du}{dx} = 6x^5$$

$$\frac{df}{dx} = \frac{du}{dx} / \frac{dv}{dx} = \frac{3x^2}{6x^5} = \frac{1}{2}x^{-3}$$

Function $y = f(x)$	**Derivative $\frac{dy}{dx}$**
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The Quotient Rule

- This is incorrect as from the table of common differentials we know that

$$\frac{df}{dx} = -3x^{-4}$$

- When we have a functions dividing another function, $f(x) = \frac{u(x)}{v(x)}$, we can use the **quotient rule**:

$$\frac{df}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

- This rule allows us to increase the complexity of the functions we study!

Function $y = f(x)$	**Derivative $\frac{dy}{dx}$**
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x^n	nx^{n-1}
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e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
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The Quotient Rule

- Let's see it in action

$$f(x) = \frac{3x+1}{6x-1}$$

- **Step 1**: Split the function:

$$u(x) = 3x+1 \qquad v(x) = 6x-1$$

- **Step 2**: Differentiate:

$$\frac{du}{dx} = 3 \qquad \frac{dv}{dx} = 6$$

- **Step 3**: Apply the product rule and substitute back in:

$$\frac{df}{dx} = \frac{3(6x-1) - 6(3x+1)}{(6x-1)^2} = -\frac{9}{(6x-1)^2}$$

The Quotient Rule

$$\frac{df}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

Function $y = f(x)$	Derivative $\frac{dy}{dx}$
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